

Maternal obesity may disrupt offspring metabolism by inducing oocyte genome hyper-methylation via increased DNMTs

Reviewed Preprint

v1 • August 13, 2024

Not revised

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Abstract

Maternal obesity has deleterious effects on oocyte genome methylation establishment, yet the underlying mechanisms remain unclear. In the present study, we first find that maternal obesity induced by high-fat diet (HFD) disturbs genomic methylation in oocytes, and at least a part of the altered methylation is transmitted to F2 oocytes and livers via females. We further identified that altered metabolites such as methionine and melatonin may play a key role in the re-methylation establishment in oocytes of obese mice. Exogenous melatonin treatment significantly reduces the hyper-methylation of HFD oocytes. The higher expression of DNMT3a and DNMT1 in HFD oocytes is also decreased by melatonin supplement, which may be mediated by cAMP/PKA/CREB pathway. These results suggest that maternal obesity-induced genomic methylation alterations in oocytes, can be partly transmitted to F2 in females, and that melatonin is involved in regulating the hyper-methylation of HFD oocytes via increasing the expression of DNMTs mediated by cAMP/PKA/CREB pathway.

eLife assessment

This manuscript reports **important** findings on the impact of maternal obesity on oocyte methylation and its transgenerational effects. The evidence presented to substantiate the major claims appears **incomplete**. This study would be of interest to biologists in the fields of epigenetics and metabolism.

<https://doi.org/10.7554/eLife.97507.1.sa3>

Introduction

Obesity has become a global health problem, affecting about 13% of the world's adult population, and more than 340 million children and adolescence (WHO). Obesity has negative effects on reproduction and offspring health. It's demonstrated that maternal obesity not only reduces the function of the hypothalamic-pituitary-ovarian (HPO) axis (1), oocyte cytoplasmic quality, and nuclear maturation, but also induces abnormal genome methylation (2). Reduced expression of *Stella* in oocytes induced by obesity results in global hypo-methylation in zygotes, which is an important contribution to the defective embryo development (3). Offspring of obese females also shows a higher risk of non-communicable diseases, such as obesity, diabetes, and cardiovascular diseases (4). Our previous study demonstrates that obesity altered the methylation level of *Leptin*, which might play a role in the metabolic disorders of female offspring, but the abnormal methylation of *Leptin* is not observed in F1 oocytes (5). The influence of maternal obesity on genomic methylation of oocytes is still obscure. Therefore, more studies are necessary to explore the role of methylation in mediating the transgenerational transmission of metabolic syndrome induced by female obesity.

Obesity results in abnormalities of glucose and lipid metabolism, which has negative effects on oocyte maturation and embryo development. In obese females, the circulating free fatty acids level is higher, which induces lipotoxicity to oocytes. The mitochondrial function of oocytes from obese females is also decreased (2). These indicate that the disturbed metabolism induced by obesity plays a key role in the reduced oocyte quality. Therefore, we suppose that disturbed metabolism induced by obesity may be an important reason for the abnormal methylation in oocytes.

In the present study, we investigated the effects of metabolites in obese mice on genomic methylation of oocytes and its inheritance via females.

Results

Obesity alters genomic methylation in oocytes

Obese mouse model was induced by feeding high-fat diet (HFD) (3, 5), and mice fed with normal diet was used as a control (CD). The average body weight of HFD group was significantly higher than that in the CD group ($n > 86$, Fig. S1A and B). Re-methylation in oocytes occurs in follicular development and is nearly completed at the germinal vesical stage (GV). The 5mC (5-methylcytosine) and 5hmC (5-hydroxymethylcytosine) levels in the GV oocytes of HFD mice were significantly higher than in the CD group ($n > 30$, Fig. 1A-C). To further explore the effects of maternal obesity on oocyte methylation, we examined the genomic methylation of metaphase II (MII) oocytes using whole genome bisulfite sequencing for small samples (WGBS, Novogene, Beijing, China). Results showed that the global methylation in MII oocytes of HFD group was higher than that in the CD group (Fig. 1D). Methylated cytosine (C) showed three types in the genome including CG, CHG and CHH (H=A, T or C), and methylated CG had more contribution in regulating gene expression. Therefore, we further analyzed the CG methylation in the present study. We found that the CG methylation level in HFD MII oocytes was significantly higher than that in the CD group (Fig. 1E). Differentially methylated CG distributed at all chromosomes (Fig. S1C). To further analyze the distribution of methylation, each function region of gene was equally divided into 20 bins, and then the average methylation levels of function regions of all genes were calculated, respectively. CGIs (CG islands) and CGI shore were predicted using cpGISlandExt and

repeat sequences were predicted using RepeatMasker. Results showed that hyper-methylation was mainly distributed at promoter, exon, upstream 2k, and downstream 2k regions (**Fig. 1F** and **Fig. S1D**) in oocytes of obese mice.

Distribution of differentially methylated regions (DMRs)

We further analyzed the differentially methylated regions (DMRs) in oocytes of both HFD and CD groups, and identified 4340 DMRs at the standards: the number of CG ≥ 4 and the absolute methylation difference ≥ 0.2 , including 2013 hyper-DMRs (46.38%) and 2327 hypo-DMRs (53.62%) (**Fig. 1G**), which distributed at all chromosomes (**Fig. 1H**). We then annotated DMRs into different genomic regions including promoter, exon, intron, CGI, CGI shore, repeat, TSS (transcription start site), TES (transcription end site), UTR3 (3' end untranslated region), and UTR5 (5' end untranslated region) (**Fig. S2A**), and the average methylation level of DMRs in these regions were similar between HFD and CD oocytes (**Fig. S2B**). Methylation level at promoters has great contributions in regulating gene expression. We then analyzed the enrichment of genes with DMRs at promoters in KEGG (Kyoto Encyclopedia of Genes and Genomes) pathways using KOBAS online. Results indicated that genes with DMRs at promoters were significantly enriched in metabolic pathways including amino acid metabolism pathways, carbohydrate metabolism pathways, lipid metabolism pathways, metabolism of cofactors and vitamins pathways, and so on (**Fig. 1I**, **Table S1**). A total of 35 genes with DMRs at promoters were included in metabolism pathways, and 19 genes were with hyper-DMRs and 16 genes were with hypo-DMRs (**Table S2**). These results suggest that the altered methylation in oocytes induced by maternal obesity may play a role in the offspring metabolic disorders.

Disturbed methylation can be inherited transgenerationally through females

Both oocyte quality and uterine environment have contributions to adult diseases that may be mediated by epigenetic modifications (6). Our recent study demonstrates that disturbed methylation in oocytes caused by uterine undernourishment can be partly transmitted to F2 oocytes via female, which may play a key role in the transgenerational inheritance of metabolic disorders (7). To investigate the inheritance of altered methylation in oocytes, we bred F1 and F2 generations as shown in **Fig. 1J**: F1, female HFD (HF1) and CD (CF1) respectively mated with control male; F2, female HF1 (HF2) and CF1 (CF2) respectively mated with control male. We examined the glucose and insulin tolerance (GTT and ITT), and results showed that the impaired GTT and ITT w in F0, F1, and F2 females (**Fig. 2A-C**).

To confirm the transmission of disturbed methylation via females, we examined the methylation status of some DMRs located at promoters of *Bhlha15* (also known as *Mist1*, basic helix-loop-helix, a transcriptional factor), *Mgat1* (mannoside acetylglucosaminyltransferase 1), *Taok3* (Serine/threonine-protein kinase 3), *Tkt* (transketolase), *Pik3cd* (phosphatidylinositol-4, 5-bisphosphate 3-kinase catalytic subunit delta), and *Pld1* (phospholipase D1). We found that the methylation levels of hyper-DMRs including *Bhlha15*-DMR, *Mgat1*-DMR, and *Taok3*-DMR in HFD oocytes were significantly higher than in the CD group (**Fig. 2D**); the methylation level of hypo-DMRs *Pik3cd*-DMR in HFD oocytes was significantly lower than that in the CD group (**Fig. 2D**). These results coincide with the genomic sequencing results. But the methylation level of *Tkt*-DMR (hypo-DMR) in HFD oocytes was higher than that in the CD group, and the methylation level of *Pld1*-DMR (hypo-DMR) was similar between two groups, which were contradictory with the genomic sequencing results (**Fig. S3A**). These results suggest that some regions are false-positive in genomic sequencing result. To exclude the effects of somatic cell contamination, we examined the methylation level of paternally imprinted gene *H19*, and it was low in both HFD and CD oocytes (**Fig. S3B**). This indicates that samples are not contaminated by somatic cells.

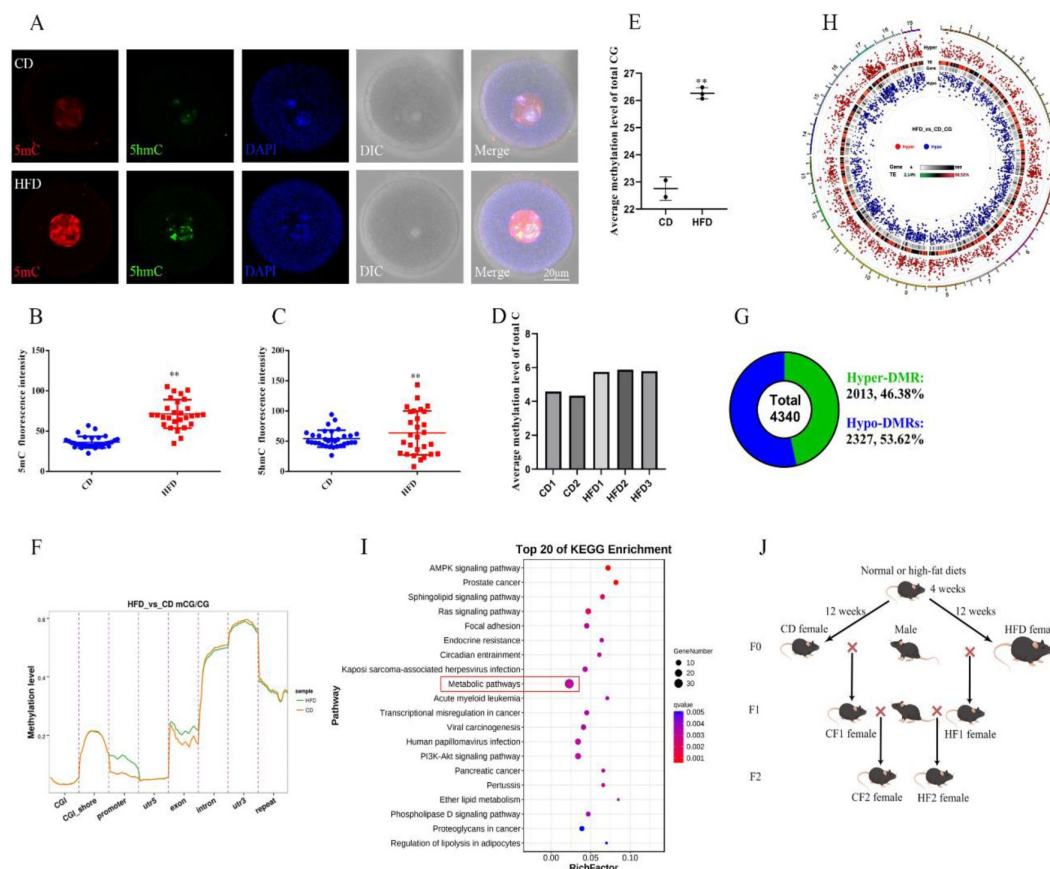


Figure 1

Maternal obesity alters DNA methylation of oocytes.

(A) Methylation level of 5mC and 5hmC in oocytes. 5mC, 5-methylcytosine; 5hmC, 5-hydroxymethylcytosine; DAPI, chromatin. (B, C) Relative fluorescence intensity of 5mC and 5hmC in GV oocytes. (D) Genomic methylation level of MII oocytes examined by single-cell whole genome bisulfite sequencing. Control group (CD) has two replicates, and obesity group (HFD) has three replicates. (E) Average genomic CG methylation level in MII oocytes. CD, control group; HFD, obesity group; ** means p value < 0.01 . (F) CG methylation levels at different regions in MII oocytes. CGI, CpG Island; utr5, 5' untranslated region; utr3, 3' untranslated region; repeat, repeat sequence. (G) Total differentially methylated regions (DMRs) in oocytes of control and obesity groups. hyper-DMRs, hypermethylated DMRs; hypo-DMRs, hypomethylated DMRs. (H) Distribution of DMRs on chromosomes in MII oocytes. Outside-to-in: chromosome, hyper-DMRs, TE (transcription end region), and gene, hypo-DMRs. (I) Enrichment of genes with DMRs at the promoter regions in KEGG pathways, and the top 20 pathways were presented. (J) Schedule of breeding. Female C57BL/6 were fed with normal (CD) or high-fat diet (HFD) for 12 weeks marked as F0. F1 was produced by F0 mating with normal males, respectively, and marked as CF1 and HF1; F2 was produced by female F1 mating with normal males and marked as CF2 and HF2, respectively.

If the altered methylation in oocytes induced by obesity would transmit to offspring, it would be found in tissues. We then examined the methylation status of DMRs in F1 livers using bisulfite sequencing (BS). Ten livers from five litters were analyzed for each group. Results showed that the methylation levels of *Bhlha15*-DMR and *Mgat1*-DMR were higher and the methylation of *Pik3cd*-DMR was lower in livers of HF1 (Fig. S4A-C), which coincides with that in HFD oocytes. The methylation level of *Tkt*-DMR in HF1 livers was lower than that in CF1 (Fig. S4D), although it was higher in HFD oocytes. This indicates that the uterine environment of obesity and reprogramming may also have contributions to this contradiction. We further examined the expression of genes with DMRs at promoters, including hyper-methylated genes, *Bhlha15*, *Mgat1*, *Dgka*, *Pdpk1* and *Taok3*, and hypo-methylated genes, *Igf1*, *Map3k8*, *Pld1*, *Tkt*, *Pik3cd* and *Sphk2*. The expressions of *Bhlha15*, *Mgat1* and *Pdpk1* were significantly lower and the expression of *Map3k8*, *Tkt*, *Pik3cd*, and *Sphk2* were significantly higher in HF1 livers compared with CF1 group (Fig. S4E and F). The expression trends of *Bhlha15*, *Mgat1*, *Tkt*, and *Pik3cd* were coincide with the methylation status at promoters in HF1 livers. The expressions of *Dgka*, *Taok3*, *Igf1*, and *Pld1* were not affected in HF1 livers (Fig. S4E and F). *Mgat1* is associated with lipid metabolism and obesity (8, 9), *Tkt* regulates glucose metabolism (10, 11), and *Pik3cd* is involved in lipid metabolism and diabetes (12, 13). These results suggest that the altered methylation in HFD oocytes is partly transmitted to F1 livers via oocytes, and abnormal methylation may be a reason for the disturbed metabolism of HF1.

If the altered methylation in HFD oocytes were inherited by HF1 oocytes, it would be transmitted to F2 generation. Therefore, we examined the methylation of DMRs in HF1 oocytes, and results showed that *Bhlha15*-DMR, *Mgat1*-DMR, and *Taok3*-DMR were significantly hyper-methylated in HF1 oocytes compared with CF1 group (Fig. 2E). The methylation level of *Tkt*-DMR was significantly lower (Fig. S5) and the methylation level of *Pik3cd*-DMR was slightly lower in HF1 oocytes than that in CF1 group (Fig. 2E). These results indicate that at least a part of the altered methylation in HFD oocytes is transmitted to F1 oocytes via females. Therefore, we further produced F2 using female HF1 and CF1 to research the transgenerational transmission of disturbed methylation in HFD oocytes (Fig. 1J). The methylation levels of *Bhlha15*-DMR and *Mgat1*-DMR were higher and the methylation level of *Pik3cd*-DMR was lower in HF2 livers compared with CF2 group (Fig. S6A-C). The methylation of *Tkt*-DMR in HF2 livers was similar to CF2 group (Fig. S6D). In addition, the expressions of *Bhlha15*, *Pdpk1*, and *Mgat1* were significantly lower (Fig. S6E), and the expressions of *Pld1*, *Pik3cd*, and *Sphk2* were significantly higher in HF2 livers compared with CF2 group (Fig. S6F). The other genes' expressions were similar in livers between HF2 and CF2 groups (Fig. S6E and F). These results suggest that at least a part of the altered methylation in HF2 livers may be inherited from HFD oocytes, which is associated with the altered expressions of genes.

Results in Fig. S6 showed that at least a part of the altered methylation in DMRs was transmitted to F2 livers via female. Thus, we supposed that the altered methylation can be, at least partly, inherited by F2 oocytes. We then examined the methylation of DMRs in F2 oocytes using BS, and found that the hyper-methylation of *Bhlha15*-DMR, *Mgat1*-DMR, and *Taok3*-DMR, and the hypo-methylation of *Pik3cd*-DMR and *Tkt*-DMR were maintained in HF2 oocytes compared with CF2 group (Fig. 2F). These results indicate that HF2 oocytes may inherit, at least partly, the altered methylation in HFD oocytes via females.

To better understand the inheritance of the altered methylation in HFD oocytes, we systematically analyzed the methylation status in different generations (Fig. 2G). For hyper-DMRs, the hyper-methylation of *Bhlha15*-DMR was maintained from HFD oocytes to F2 oocytes, and the hyper-methylation of *Mgat1*-DMR and *Taok3*-DMR was partly transmitted to F2 from HFD oocytes. For hypo-DMRs, the hypo-methylation of *Tkt*-DMR was partly maintained from F1 to F2, and the hypo-methylation of *Pik3cd*-DMR in HFD oocytes was inherited by F2 (Fig. 2G). For *Pld1*-DMR, the methylation level was similar between CD and HFD oocytes. So we didn't further examined it in F1

and F2 (**Fig. 2G**). These results suggest that only a part of the altered methylation in HFD oocytes can be transmitted to F2 oocytes, and the disturbed methylation may be a reason for the inheritance of metabolic disorders.

Obesity alters the metabolomics of serum

Obesity alters the metabolism of glucose, fatty acid, and amino acids, which are essential for oogenesis. Thus, we suppose the altered metabolism may play a key role in the disturbed global methylation in HFD oocytes. We examined the metabolomics of serum using non-target approaches (BGI, Wuhan, China). We used LC-MS/MS to identify the variation of metabolites including amino acids, carbohydrates, lipid, phenols, and so on. The principal component analysis (PCA) showed that the PC1 and PC2, respectively, explained 44.6% and 9.96% of the total metabolite variations (**Fig. 3A**). The distribution of extracts between groups was distinguishable (**Fig. 3A**). We identified 538 differential metabolites based on the PLS-DA (robust orthogonal partial least squares-discriminant analysis) and T-test was used for analysis: VIP (variable importance in projection) ≥ 1 , Fold-change ≥ 1.2 or ≤ 0.83 , and p-value < 0.05 , including 288 up-regulated and 250 down-regulated (**Fig. 3B**). The enrichment of differential metabolites was analyzed using KEGG and found that differential metabolites were significantly enriched in tryptophan and vitamin B6 metabolism (**Fig. 3C**). The top 20 differential metabolites were presented in **Fig. 3D**. These results suggest that obesity disturbs the metabolomics of serum.

Melatonin may play a key role in the genomic hyper-methylation of HFD oocytes

To investigate the association between metabolites and methylation, we identified some differential metabolites in HFD mice, including pyridoxine (vitamin B6), L-methionine, melatonin, and L-tyrosine that may be associated with hyper-methylation of oocytes. The concentrations of pyridoxine (vitamin B6), L-methionine, and L-tyrosine were significantly increased in HFD serum compared with the CD group (**Fig. 3D-G**). In the methionine cycle, methionine adenosyltransferase catalyzes methionine and ATP into S-adenosyl methionine (SAM) (14) which acts as a universal methyl donor (15). DNA methyltransferases (DNMTs) transfer the methyl group to cytosine and convert SAM to S-adenosyl homocysteine (SAH) which is further degraded to homocysteine (HCY) by SAH hydrolase (16). During these processes, vitamin B6 serves as a co-factor (17). It's reported that excessive methionine and vitamin B6 intake induces hyper-methylation (18). This indicates that the hyper-methylation in HFD oocytes may be associated with the higher concentration of methionine and pyridoxine. To confirm this hypothesis, we examined the concentration of SAM, SAH, and HCY in livers and oocytes, which are crucial intermediate metabolites in methionine cycle. Results showed that the concentrations of SAM, SAH, and HCY in HFD livers were similar to that in CD livers (**Fig. 3H-J**), and the concentration of SAM in HFD oocytes was lower than that in the CD group (**Fig. 3K**). These results suggest that the higher concentrations of methionine and pyridoxine in serum of HFD mice may be not the main reason for the genomic hyper-methylation in oocytes.

As presented in **Fig. 3D and L**, the concentration of melatonin in HFD serum was significantly lower than that in the CD group. Low concentration of melatonin is also reported in obese rats (19) and humans (20). Exogenous melatonin supplementation reduces body weight and improves lipid and glucose metabolism in animals and humans (21). Melatonin also can inhibit cancer by regulating DNA methylation status (22), improve DNA methylation reprogramming in development of porcine cloned embryos (23), and affect DNA re-methylation in oocytes (24–26). This indicates that the reduced melatonin may contribute to the hyper-methylation of HFD oocytes. In the present study, we found that if HFD mice were treated with exogenous melatonin for 14 days, the genomic hyper-methylation in HFD oocytes was significantly reduced ($n \geq 96$ **Fig. 3M and N**). These results suggest that the reduced melatonin concentration may be involved in regulating the hyper-methylation in HFD oocytes.

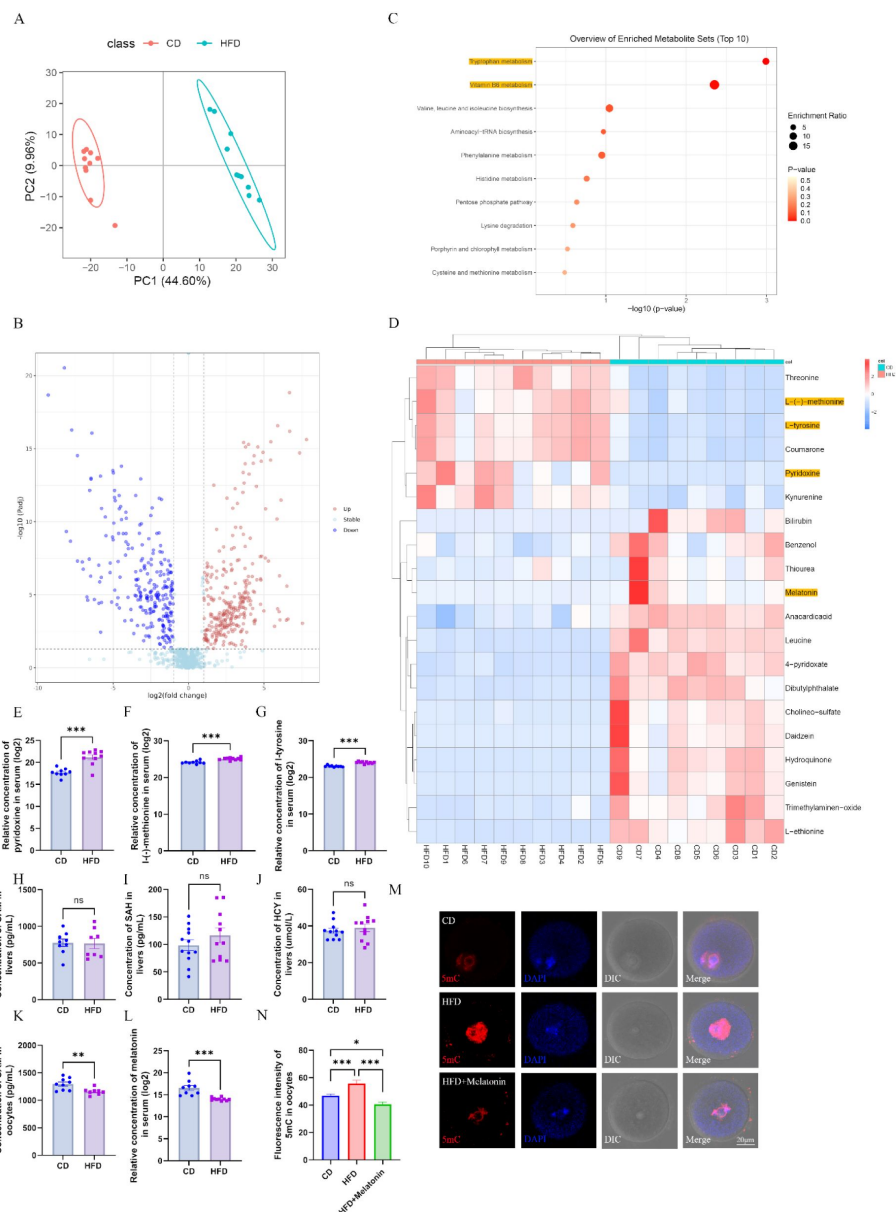


Figure 3

Maternal obesity alters metabolomics of serum.

- (A) Principal component analysis in CD and HFD mice.
- (B) Differential metabolites in HFD serum compared to CD group. Red circle, up regulated metabolites; blue circle, down regulated metabolites.
- (C) The enrichment of differential metabolites was analyzed using KEGG, and the top 10 enrichment term was presented.
- (D) Heat map of the top 20 differential metabolites in HFD serum.
- (E-G) Comparison of the concentrations of pyridoxine, methionine, and tyrosine between groups. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.
- (H-J) Concentrations of SAM, SAH and HCY in livers were examined by ELISA. Ns, there is no statistical significance between groups.
- (K) Concentration of SAM in oocytes was analyzed using ELISA. ** $p < 0.01$.
- (L) Relative concentration of melatonin in serum. *** $p < 0.001$.
- (M) Genomic DNA methylation in oocytes was examined using immunofluorescence. CD, control group; HFD, obesity group; HFD+melatonin, obese mice were treated with exogenous melatonin for 14 days.
- (N) Relative fluorescence intensity of 5mC was examined using Image J. * $p < 0.05$; *** $p < 0.001$.

Melatonin regulates genomic methylation of oocytes by increasing the expression of DNMTs via cAMP/PKA/CREB pathway

Melatonin receptors (MT1 and 2), which couples with inhibitor G-protein (Gi), are identified in oocytes and granulosa cells (27, 28). The activated Gi inhibits the activation of adenylyl cyclases (ADCYs), resulting in decrease of cAMP (cyclic adenosine monophosphate), which regulates the activation of protein kinase A (PKA) and CREB (cAMP response element-binding protein) (29). Elevated cAMP level increases the expression of DNMTs resulting in hyper-methylation in HL-1 cardiomyocytes (30). Hedrich *et al.* report that CREMa induces hyper-methylation of CD8 cluster via increasing the expression of DNMT3a (31). We thus supposed that melatonin may regulate genomic methylation in oocytes via increasing the expression of DNMTs through cAMP/PKA/CREB pathway (Fig. 4A). To confirm this hypothesis, female C57BL/6 mice fed with normal diet were treated with luzindole, an inhibitor of melatonin receptor, and the global methylation of 5mC and 5hmC was significantly increased in oocytes (n≥49 Fig. 4B-D). Luzindole didn't affect the concentration of melatonin in serum (Fig. S7A). However, excessive melatonin treatment significantly increased the concentration of melatonin in serum (Fig. S7A) and decreased the methylation level of 5mC and 5hmC in oocytes (Fig. 4B-D). These indicate that melatonin may regulate re-methylation process in oocytes.

To confirm whether melatonin regulates methylation in oocytes via cAMP pathway, we examined the expression of ADCYs in oocytes using RT-PCR, and found that ADCY5, 6, and 9 were expressed in oocytes (Fig. S7B). Melatonin antagonist luzindole significantly increased the expression of ADCY6 and ADCY9 in oocytes, and melatonin reduced the expression of ADCY6 (Fig. 4E). However, the expression of ADCY5 was lower in luzindole and melatonin groups compared with the control (Fig. 4E). In addition, melatonin antagonist luzindole increased while melatonin decreased the concentration of cAMP in oocytes, respectively (Fig. 4F). These results suggest that melatonin may regulate the synthesis of cAMP via ADCY6 in oocytes. To further confirm the role of cAMP in regulating methylation of oocytes, we treated mice with SQ22536, an inhibitor of ADCYs, and showed that this treatment significantly reduced the global methylation of 5mC and the concentration of cAMP in oocytes (n≥51 Fig. 5A-C). Whereas, the ADCYs activator forskolin significantly increased the cAMP concentration and global methylation of 5mC in oocytes (Fig. 5A-C). 8-Bromo-cAMP, a cAMP analogue, also increased the global methylation in oocytes (n≥41 Fig. 5D, E). cAMP has function by activating the downstream protein PKA. When we treated mice with H89 2HCL, a PKA antagonist, the global methylation of 5mC was significantly reduced in oocytes (n≥24 Fig. 5F, G). These results suggest that melatonin may mediate methylation of oocytes via cAMP/PKA pathway.

cAMP activates PKA which further phosphorylates CREB to regulate gene expression. In oocytes, DNA re-methylation is regulated by DNMTs including DNMT3a, DNMT3L, and DNMT1. Therefore, we next investigated whether melatonin regulates DNMTs expression via cAMP/PKA/CREB pathway in oocytes. We examined the expression of CREB1, CREM (cAMP responsive element modulator), CREB3L2 (cAMP responsive element binding protein 3 like 2), and ATF1 (activating transcription factor 1) in oocytes. Results showed that ADCYs activator forskolin treatment significantly increased the mRNA expression of CREB1 and CREM, and the expression of CREB3L2 and ATF1 was slightly increased (Fig. 6A). In addition, ADCYs inhibitor SQ22536 significantly reduced the expression of CREB1 and CREB3L2, although the expression of CREM and ATF1 was slightly decreased in oocytes (Fig. 6A). Furthermore, ADCYs inhibitor SQ22536 treatment significantly reduced the concentration of pCREB1, but it was increased by forskolin treatment in oocytes (n≥36 Fig. 6B, C). pCREB1 level was also increased by 8-Bromo-cAMP and decreased by PKA antagonist H89 2HCL in oocytes (n≥28 Fig. 6D-G). These suggest that the expression and phosphorylation of CREB1 can be regulated by cAMP/PKA pathway. Yang *et al.* demonstrate that CREB regulates DNMT3a expression in neurons of the dorsal root ganglion via binding to the promoter region (32). In the present study, the binding of pCREB1 with relative regions of

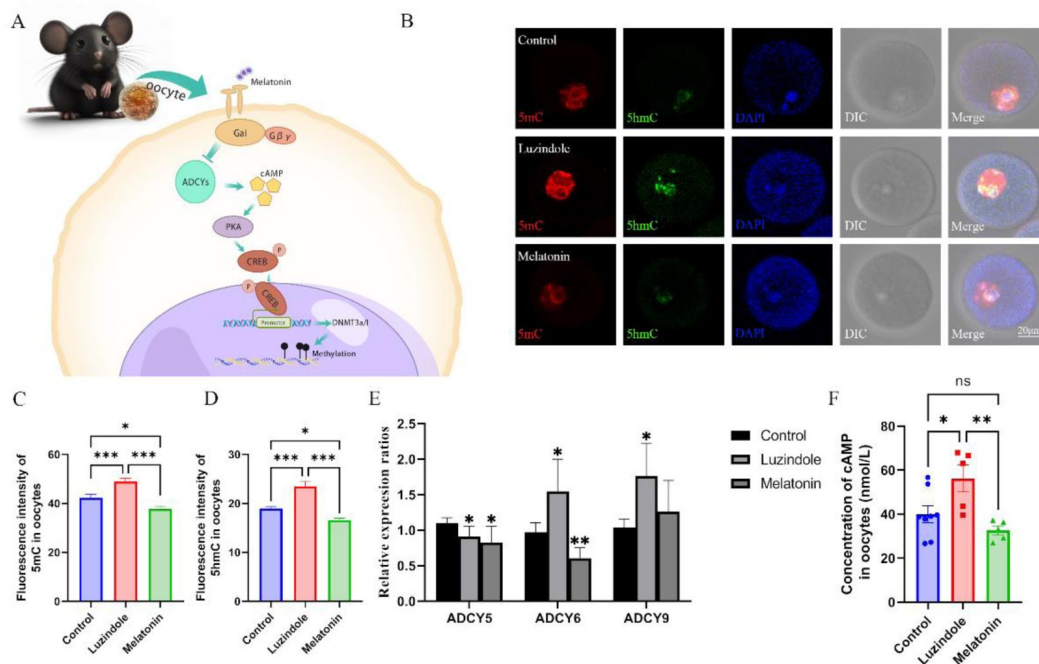


Figure 4

Melatonin regulates DNA methylation in oocytes.

(A) Schedule of possible pathway of melatonin regulating DNA methylation in oocytes. According to previous studies, we predicted that melatonin might regulate DNA methylation in oocytes via cAMP/PKA/CREB pathway.

(B) Effects of melatonin and its inhibitor luzindole on oocyte methylation were examined using immunofluorescence.

(C, D) The relative fluorescence intensities of 5mC and 5hmC were analyzed using Image J. * $p < 0.05$; *** $p < 0.001$.

(E) The effects of melatonin and its inhibitor luzindole on the expression of adenylate cyclase (ADCYs) in oocytes were examined by qPCR. * $p < 0.05$; ** $p < 0.01$.

(F) Concentration of cAMP in oocytes was examined by ELISA. * $p < 0.05$; ** $p < 0.01$.

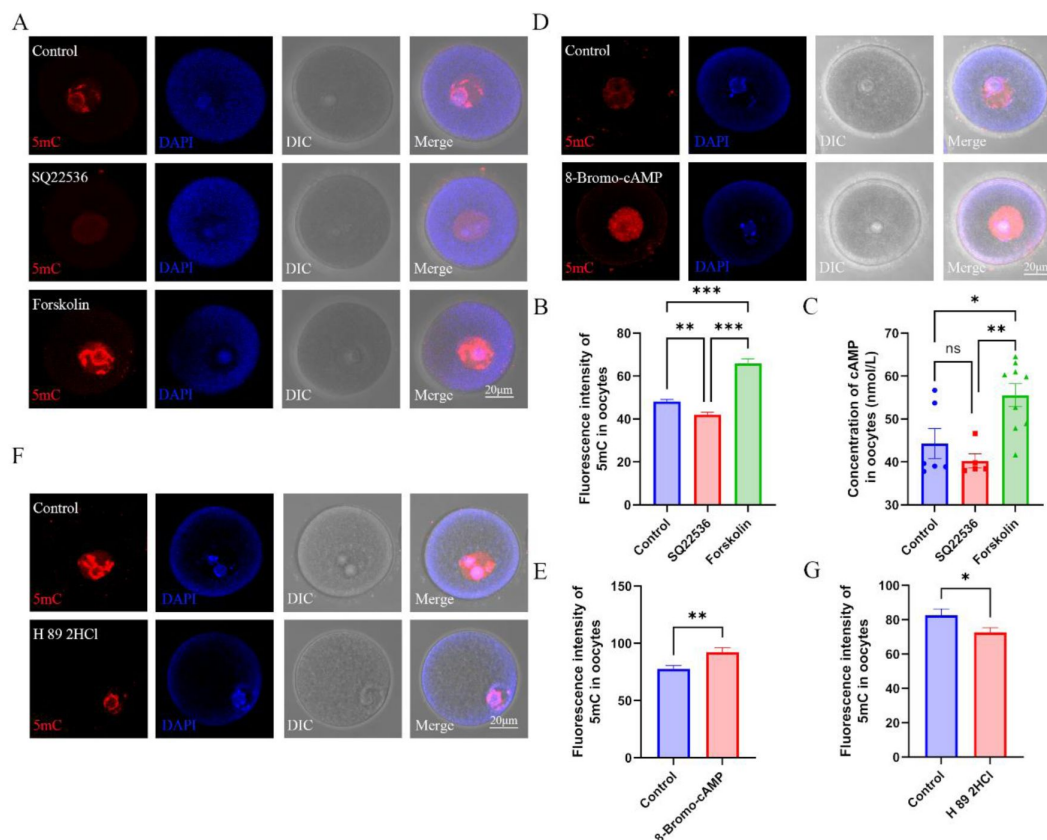


Figure 5

Role of cAMP in DNA methylation in oocytes.

(A) Female mice were respectively treated with ADCYs inhibitor SQ22536 or activator forskolin. Oocyte methylation was examined using immunofluorescence.

(B) The relative intensity of fluorescence in oocytes was analyzed using Image J. ** p < 0.01; *** p < 0.001.

(C) cAMP concentration in oocytes was examined using ELISA. * p < 0.05; ** p < 0.01.

(D) Female mice were treated with cAMP analogue 8-Bromo-cAMP, and oocyte methylation was examined using immunofluorescence. The relative fluorescence intensity of 5mC was analyzed using Image J (E). ** p < 0.01. (F, G) Female mice were treated with PKA (protein kinase A) antagonist H 89 2HCL, and then oocyte methylation was examined using immunofluorescence. The relative fluorescence intensity of 5mC was analyzed using Image J (G). * p < 0.05.

DNMTs was examined using CUT & Tag assay. Each sample contained 500 GV oocytes, and two replicates were involved. Sequencing result revealed that five fragments including 10 pCREB1 binding motifs (predicted using online tool JASPAR, Table S3) were associated with DNMTs, including 3 fragments at intron 1 and distill intergenic regions of DNMT3a, 1 fragment at promoter region of DNMT1, and 1 fragment at intron 13 of DNMT3l (Table S4). These results suggest that pCREB1 may have contributions to regulate the expression of DNMTs.

Next, we investigated the expression of DNMT1, DNMT3a, and DNMT3l in oocytes. Melatonin inhibitor luzindole (slightly) and ADCYs activator forskolin (significantly) increased the expression of DNMT1 and DNMT3a, respectively (Fig. 7A, B). Melatonin and ADCYs inhibitor SQ22536 significantly and slightly reduced the expression of DNMT1 and DNMT3a in oocytes, respectively (Fig. 7A, B). The protein level of DNMT3a in GV oocytes was also significantly increased by 8-Bromo-cAMP and decreased by PKA antagonist H 89 2HCL, respectively (n≥48 Fig. 7C-F). Although DNMT1 is well known as maintenance methyltransferase, it also has contribution to *de novo* methylation in oocytes (33). Therefore, we examined the localization of DNMT1 in oocytes, and found that 8-Bromo-cAMP treatment significantly increased the localization of DNMT1 in the nucleus of oocytes, but it was reduced by PKA antagonist H89 2HCL (n≥22 Fig. 7G-J). When the activation of DNMTs was inhibited by 5-azacytidine, the methylation level in GV oocytes was significantly decreased (n≥43 Fig. S8). These results suggest that melatonin may influence the genomic methylation of oocytes via regulating the expression of DNMT1 and DNMT3a mediated by cAMP/PKA/CREB pathway.

Increased DNMTs mediates hyper-methylation of HFD oocytes via cAMP/PKA/CREB pathway

To explore how melatonin regulates the hyper-methylation of HFD oocytes, we examined the expression of DNMTs. Results showed that maternal obesity in mice significantly increased the expression of DNMT1, DNMT3a, and DNMT3l in oocytes (Fig. 8A). So, we would like to confirm whether cAMP/PKA/CREB pathway mediated the increased expression of DNMTs in HFD oocytes. We examined the concentration of cAMP in oocytes, and found that maternal obesity significantly increased the concentration of cAMP in oocytes compared with CD group (Fig. 8B). The mRNA expression of CREB1, but not CREM, in HFD oocytes was significantly increased compared with CD group (Fig. 8C), and the pCREB1 level in HFD oocytes was also significantly increased (n≥49 Fig. 8D, E). But the higher level of pCREB1 was reduced by exogenous melatonin treatment (Fig. 8D, E). When obese females were treated with PKA antagonist H89 2HCL, both 5mC and pCREB1 levels were significantly reduced in oocytes (n≥17 Fig. 8F-I). These results suggest that reduced melatonin in obese mice may increase the expression of DNMTs by cAMP/PKA/CREB pathway.

The higher expression of DNMT3a (Fig. 8A) may have contribution to the genomic hyper-methylation of HFD oocytes because the primary function of DNMT3a is *de novo* DNA methylations. The *de novo* methylation function of DNMT1 is usually prevented by *Stella* (also known as *Dppa3* or *PGC7*) in oocytes (34). Nevertheless, maternal obesity significantly reduces the expression of *Stella* in oocytes (3), which indicates that DNMT1 may also have contribution to the hyper-methylation of HFD oocytes. We found that the localization of DNMT1 in HFD oocyte nucleus was significantly increased which could be reduced by PKA antagonist H89 2HCL treatment (n≥24 Fig. 8J, K). These results suggest that increased DNMTs induced by maternal obesity has contributions to the hyper-methylation of oocytes.

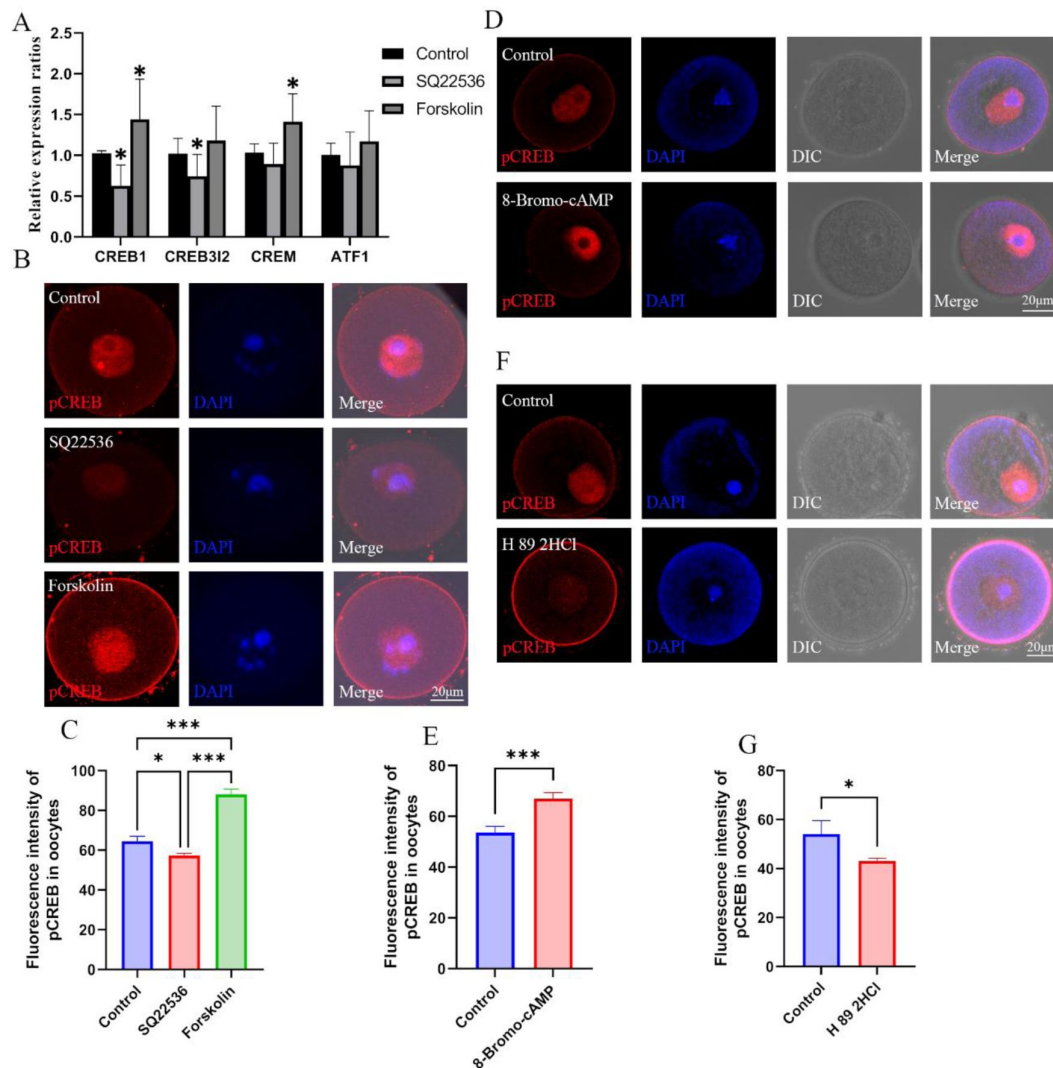


Figure 6

Effects of cAMP on CREB1.

(A) The mRNA expression of cAMP-response element binding (CREB) proteins in oocytes was examined by qPCR. * $p < 0.05$. (B and C) Phosphorylated CREB1 (pCREB1) in oocytes was examined using immunofluorescence, and the relative fluorescence intensity of pCREB1 was examined by Image J (C). * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. (D and E) After treatment of cAMP analogue 8-Bromo-cAMP, pCREB1 in oocytes was examined using immunofluorescence. The relative fluorescence intensity was analyzed using Image J (E). *** $p < 0.001$.

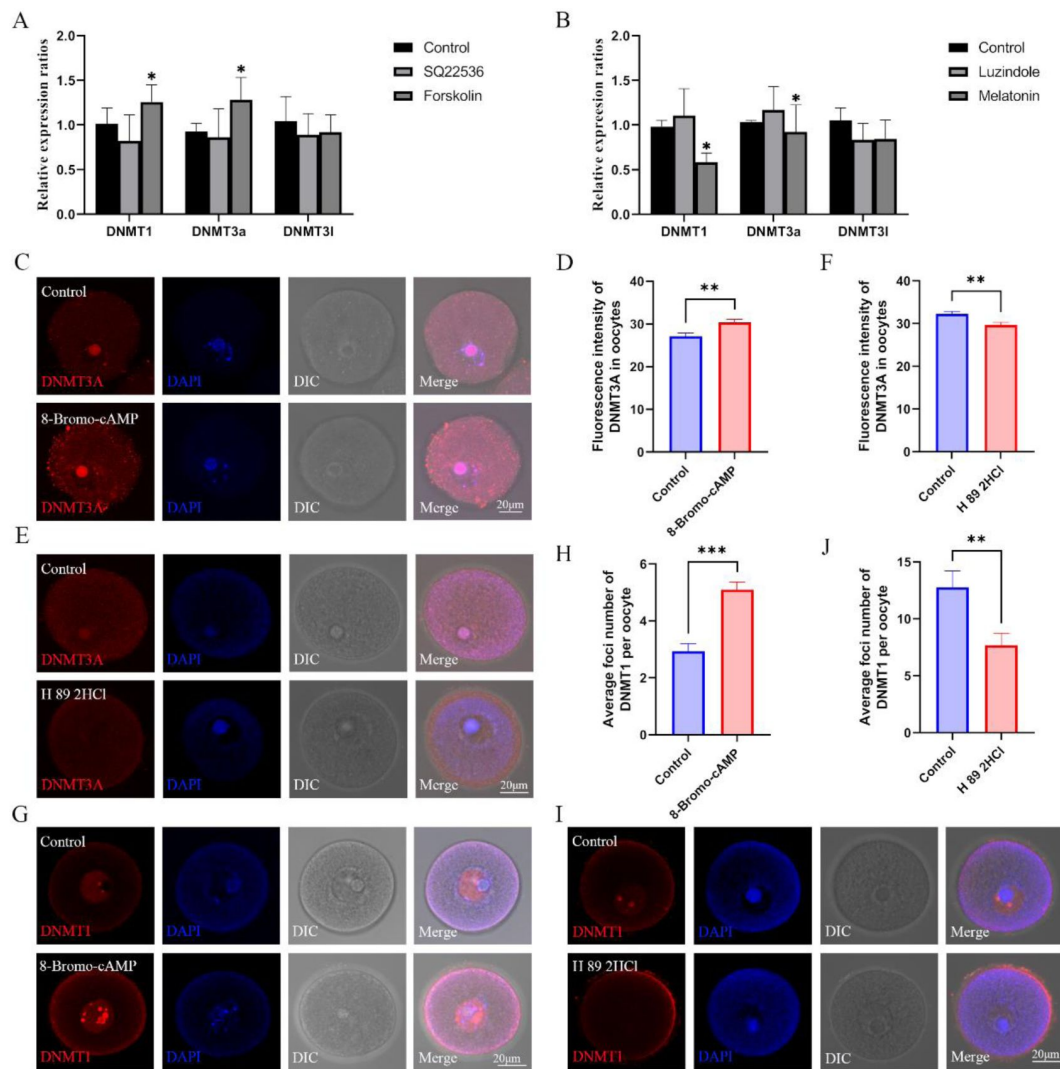


Figure 7

Role of melatonin/cAMP/PKA pathway in the expression of DNMTs.

(A) The expressions of DNMT1, DNMT3a and DNMT3l in oocytes were examined using qPCR after the treatment with SQ22536 and forskolin. * $p < 0.05$.

(B) Relative expressions of DNMT1, DNMT3a and DNMT3l in oocytes were examined using qPCR after the treatment with luzindole and melatonin. * $p < 0.05$.

(C) After 8-Bromo-cAMP treatment, the relative expression of DNMT3a in oocytes was examined using immunofluorescence and calculated by Image J (D). ** $p < 0.01$.

(E and F) PKA antagonist H 89 2HCL treatment significantly reduced the level of DNMT3a in oocytes examined using immunofluorescence. ** $p < 0.01$.

(G and H) DNMT1 localization in oocyte nucleus was examined using immunofluorescence after 8-Bromo-cAMP treatment. *** $p < 0.001$.

(I and J) The localization of DNMT1 in oocyte nucleus was reduced by the treatment of PKA antagonist H 89 2HCL. ** $p < 0.01$.

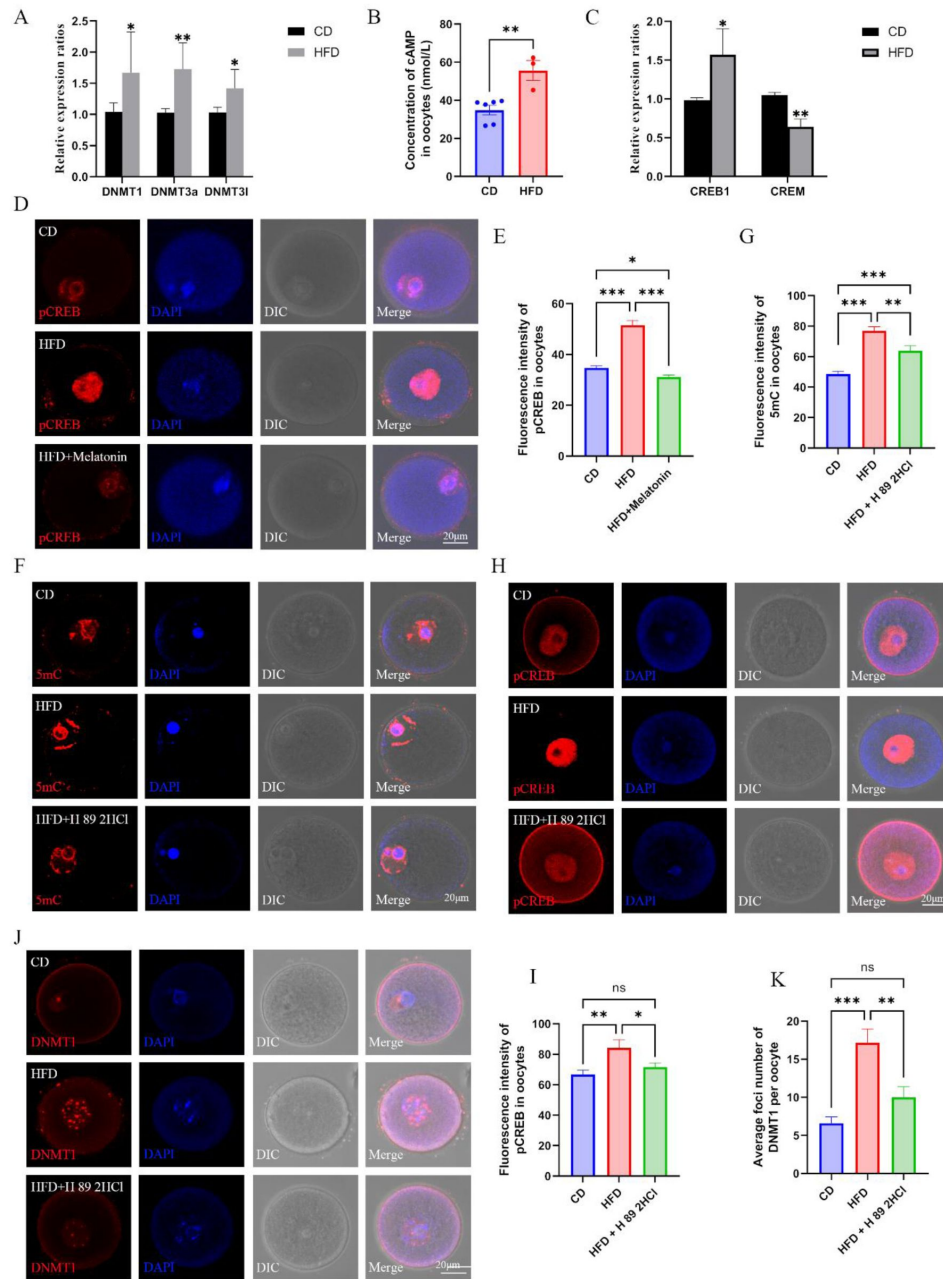


Figure 8

Melatonin regulates DNMTs expression via cAMP/PKA/CREB pathway in HFD oocytes.

(A) Relative expression of DNMT1, DNMT3a, and DNMT3I in HFD oocytes was examined using qPCR. * p<0.05; ** p<0.01.

(B) Concentration of cAMP in HFD oocytes was examined using ELISA. ** p<0.01.

(C) Relative expressions CREB1 and CREM in HFD oocytes were tested using qPCR. * p<0.05; ** p<0.01.

(D and E) The level of pCREB in oocytes was examined using immunofluorescence, and the relative fluorescence intensity was calculated by Image J (E). HFD, oocytes from obese mice; CD, oocytes from control mice; HFD + melatonin, oocytes from obese mice treated with exogenous melatonin. * p<0.05; *** p<0.001.

(F and G) PKA antagonist H89 2HCL treatment reduced the methylation level of HFD oocytes. ** p<0.01; *** p<0.001.

(H and I) The level of pCREB1 in HFD oocytes was also decreased by the treatment of PKA antagonist H89 2HCL. * p<0.05; ** p<0.01; ns, no statistical significance between groups.

(J and K) PKA antagonist H89 2HCL treatment reduced the localization of DNMT1 in HFD oocytes. ** p<0.01; *** p<0.001; ns, no statistical significance between groups.

Discussion

Maternal obesity has negative effects on oocyte quality and offspring health, but the mechanisms are not well elucidated. In the present study, we found that maternal obesity induced hyper-methylation in oocytes, and the abnormal methylation, at least partly, is transmitted to F2 oocytes in females, which may be associated with the occurrence and inheritance of metabolic disorders. Maternal obesity-induced metabolic changes may be the cause of DNA hyper-methylation, and decreased melatonin is involved in regulating the hyper-methylation of HFD oocytes via increasing the expression of DNMTs mediated by cAMP/PKA/CREB pathway.

Transgenerational epigenetic inheritance is common in plants, but related investigation in mammals is hindered by epigenetic reprogramming events during gametogenesis and early embryo development (35, 36). In mammals, isogenic agouti viable yellow (A^{vy}) and axin-fused ($Axin^{Fu}$) mice, whose phenotype is regulated by the DNA methylation level of *Iap* (intracisternal A particle long terminal repeat) respectively located at the upstream and in intron 6, are solid evidence confirming the transgenerational inheritance of epigenetic modifications (37, 38). Nevertheless, epigenetic modifications can be affected by environmental factors such as metabolic diseases and diets. The intergenerational inheritance of phenotype and epigenetic changes induced by maternal environmental factors is confirmed by previous studies (5, 39), but there are still many debates about the transgenerational inheritance of epigenetic changes induced by environment. Rats from stressed mother are more likely to be stressed, and this can be transmitted across generations, but this transgenerational inheritance is not mediated by gametes (40). Females cannot mediate the transgenerational inheritance of hyper-methylation induced by diet in A^{vy} mice (41), but another study reported that if A^{vy} allele is from father, the hyper-methylation induced by diet during pregnancy can be retained in germ cells (42). Anway et al. report that the abnormal spermatogenesis induced by exposure of vinclozolin during pregnancy can be transmitted across four generations via sperm (43). We previously reported that maternal obesity disturbed DNA methylation status of imprinted genes in oocytes (5), but the transgenerational inheritance was not observed. We also demonstrate that disturbed methylation in oocytes induced by undernourishment in utero could be inherited, at least partly, by F2 oocytes via females (44). Recently, Takahashi *et al.* edited DNA methylation of promoter-associated CGIs, and found that the edited DNA methylation, associated with disturbed metabolism, was stably inherited by multiple generations (45). In the present study, we find that maternal obesity induces genomic hyper-methylation in oocytes, and a part of the abnormal methylation transmits to F2 via female gametes. Meanwhile, the transmission of metabolic disorders is also observed across two generations. These suggest that the transgenerational inheritance of abnormal methylation induced by maternal obesity, at least partly, can be mediated by oocytes, which may be a reason for the inheritance of metabolic disorders.

During methylation process, the methyl group is donated by SAM which is generated from homocysteine, 5-methyltetrahydrofolate (5mTHF), and methionine. 5mTHF is an intermediate of one-carbon metabolism (46). It's demonstrated that one-carbon units such as folate and vitamin B12 are crucial for the establishment of methylation (46). Disturbed glucose and lipid metabolism also has negative influence on DNA methylation (47). These indicate that abnormal metabolism induced by maternal obesity (48) may play a key role in the genomic hyper-methylation in oocytes. In the present study, we find that the metabolomics of serum in HFD mice is distinguishable with that in the CD control. Although the concentrations of vitamin B6 and methionine are higher in HFD serum, it may be not an important reason for the genomic hyper-methylation in oocytes because the concentrations of SAM, SAH, and HCY in livers and oocytes are similar between CD and HFD mice. In humans, obesity reduces the melatonin level in circulation (19, 20). In the present study, we also find that maternal obesity induced by high-fat diet reduces the concentration of melatonin in serum. Melatonin not only can decrease body weight,

but also regulates DNA methylation of somatic cells and germ cells (22 [↗](#), 26 [↗](#)). Nevertheless, the molecular mechanism of melatonin regulating DNA methylation in oocytes is still covered. Melatonin has two receptors, MT1 and MT2, both of which have been identified in oocytes (27 [↗](#)). Melatonin receptors couple with inhibitor G-protein and can regulate gene expression via cAMP/PKA/CREB pathway (29 [↗](#)). CREM and cAMP mediate DNA methylation in somatic cells by regulating the expression of *DNMT3a* (30 [↗](#), 31 [↗](#)). In neurons, CREB interacts with promoter of *DNMT3a* to regulate its expression and DNA methylation (32 [↗](#)). In the present study, we find that melatonin, mediated by cAMP/PKA/CREB pathway, regulates methylation in oocytes by increasing the expression of DNMT1 and DNMT3a. Similar results are also observed in HFD oocytes, and the hyper-methylation of HFD oocytes can be reduced by exogenous melatonin and PKA inhibitor. These suggest that decreased melatonin level is involved in regulating the genomic hyper-methylation of HFD oocytes via increasing the expression of DNMTs, mediated by cAMP/PKA/CREB pathway.

During follicular development, re-methylation in oocytes is mainly catalyzed by DNMT3a and DNMT3L (49 [↗](#)). DNMT1 is usually responsible for DNA methylation maintain, but DNMT1 also contributes to CG methylation in oocytes (50 [↗](#)). In normal oocyte developing process, DNMT1 is mainly prevented in the nucleus by *Stella*. While *Stella* level is reduced, *Uhrf1* (Ubiquitin-like containing PHD Ring Finger 1) moves to the nucleus from cytoplasm and recruits DNMT1 to chromatin, resulting in hyper-methylation (33 [↗](#)). Maternal obesity significantly decreases the expression of *Stella* in oocytes (3 [↗](#)). We find that the expression of DNMT1 and its localization at chromatin in oocytes are increased by maternal obesity. These suggest that reduced *Stella* in HFD oocytes may recruit more DNMT1 into chromatin resulting in hyper-methylation.

In summary, we find that maternal obesity induces genomic hyper-methylation in oocytes and at least a part of the altered methylation can be transmitted to F2 oocytes, which may be a reason for the inheritance of metabolic disorders. Furthermore, reduced melatonin in HFD mice is involved in regulating the genomic hyper-methylation of oocytes via increasing the expression of DNMTs, and this process is mediated by cAMP/PKA/CREB pathway. However, there are some limitations about the present study: there is not enough evidence to confirm the role of altered DNA methylation in metabolic disorders in offspring of obese mothers, the molecular mechanisms by which DNA methylation escapes reprogramming in oogenesis is not elucidated. There may be other mechanisms involving in regulating genomic hyper-methylation in HFD oocytes. Therefore, more studies are required in the short future.

Materials and methods

Mice

C57BL/6 mice were purchased from Jinan Pengyue Company (Jinan, China). Mice were housed in the controlled room with 12h light and 12h dark cycle, and at 23-25 °C. The Animal Ethic Committee of Qingdao Agricultural University supported all procedures (QAU201900326).

For obesity model, female C57BL/6 mice at the age of 4 weeks were randomly divided into two groups fed with high-fat diet (HFD, Research Diets, D12492, USA) and normal diet (CD) for 12 weeks, respectively. We examined the body weight every week.

Offspring were produced as the schedule **Fig.1J** [↗](#). For F1 offspring, female HFD and CD mice were mated with normal adult male C57BL/6 mice, respectively, and the offspring was marked as HF1 and CF1. To avoid the effects of males on methylation, the same males were used to produce F2: female HF1 and CF1 mating with normal males, marked as HF2 and CF2.

Immunofluorescence

Briefly, oocytes were fixed with 4% PFA (paraformaldehyde), permeabilized with 0.5% TritonX-100, and blocked with 1% BSA (bovine serum albumin). After that, oocytes were incubated with primary antibodies overnight. The secondary antibodies were stained for 1h at room temperature. Fluorescence signal was examined using a laser scanning confocal microscope (Leica SP5, Germany). Relative fluorescence intensity was examined using Image J.

Antibodies

Primary antibodies used in the present study included anti-5mC antibody (Abcam, ab73938), anti-5hmC antibody (Abcam, ab214728), anti-pCREB antibody (Cell Signaling Technology, 9198S), anti-DNMT3a antibody (Active motif, 61478), and anti-DNMT1 antibody (Active motif, 39204).

Single cell whole genome bisulfite sequencing (scWGBS) and analysis

Metaphase II (MII) oocytes were collected from the oviduct. For each sample, 100 oocytes from at least 10 mice were pooled together and transferred to lysis buffer. Genomic DNA was fragmented and the end was repaired. Then, fragmentations were ligated with adapter. Bisulfite treatment was performed using EZ DNA Methylation-Direct (Zymo Research). Lambda DNA was used as control. After that, the sequencing library was established and sequenced using Illumina HiSeq/NovaSeq (Novogene, China). Raw data quality was evaluated using FastQC, and low quality data and adapter was trimmed using fastp. Clean data was compared to the reference genome mm10. Methylated C sites calling were performed using Bismark. Differentially methylated regions (DMRs) were identified using DSS-single. The enrichment of genes in KEGG pathway was carried out using the online tool KOBAS.

qPCR

Total RNA was extracted from oocytes or tissues using RNAprep Pure Micro Kit (Tiangen, DP420) or RNA Easy Fast Kit (Tiangen, DP451). cDNA was synthesized using Hifair III 1st Strand cDNA Synthesis Kit (Yeasen, China). cDNA was used as templates to examine the relative expression of genes. Housekeeping genes *Ppia* and *Gapdh* were used as reference. Relative expression was calculated as $2^{-\Delta\Delta Ct}$.

Bisulfite sequencing (BS)

Each sample includes 5 oocytes and at least 20 samples were used for each DMR. Samples were treated as previous study ([5](#)). Briefly, samples were treated with lysis buffer and 0.3M NaOH, respectively. After that, samples were embedded with 2% low melting point agarose (Sigma), which was treated with fresh bisulfite solution (2.5 M sodium metabisulfite, Merck; 125 mM hydroquinone, Sigma; pH 5) for 4h. Treated DNA was used as template to amplify the target fragment using nest-PCR. PCR products were cloned to T-vector and sequenced. Methylation status was analyzed using BiqAnalyzer, which can remove the low converted rate (<97%) and possible repeat sequences. At least 10 available clones were used for each DMR.

Glucose and insulin tolerance

Glucose and insulin tolerance (GTT and ITT) were examined as previously reported ([7](#)). Briefly, mice were treated with glucose at 2g/kg body weight or insulin (Actrapid®, Novo Nordisk) at 10 IU/kg body weight after 16h or 4h fasting, respectively. After that, blood glucose was measured by tail blood at 0, 30, 60, 90, and 120 min, respectively.

ELISA

Concentrations of cAMP, SAM, SAH, and HCY were examined using ELISA kits (Jinma Biotechnology Co. LTD, Shanghai, China) according to the handbook. Standard curve was produced using four-parameters logistics.

Non-target metabolomics in serum

Metabolites in serum were examined using LC-MS/MS (BGI, Wuhan, China). Raw data was treated with Compound Discoverer 3.1 (Thermo Fisher Scientific, USA). After that, preprocessing of the exported data was performed using metaX. Metabolites were identified according to the databases of BMDB (BGI), mzCloud and ChemSpider (HMDB, KEGG, LipidMaps). Identified metabolites were annotated according to KEGG and HMDB. Differential metabolites were scanned using PCA and PLS-DA combined with Fold-change and Student's *t* test.

Chemicals

Inhibitors used in the present study include luzindole (Sigma), SQ22536 (Selleck), forskolin (Selleck), H89 2HCL (Selleck), and azacitidine (Selleck). 8-Bromo-cAMP was purchased from Selleck. Melatonin (Sigma) was injected by tail vein for 14 days. The treatments of the other chemicals were performed by intraperitoneal injection for 14 days. The control group was injected with relative solutions.

CUT & Tag and sequencing

For each sample, 500 oocytes were pooled together for this experiment, and two replicates were performed. Library was established using Hyperactive® Universal CUT&Tag Assay Kit (Vazyme, China) according to the manufacturer's instruction. Briefly, oocytes washed with washing buffer and transferred into tubes with ConA Beads Pro for 10min in room temperature. Putting tubes on magnetic rack for 2min, and then discarded the supernatant. After that, added 50 µl precooled antibody buffer and primary antibody overnight at 4 °C. Then, tubes were put on magnetic rack for 2 min, discarded supernatant, and added 50 µl Dig-wash buffer with secondary antibody (1:100) into tubes incubated at room temperature for 60 min. Samples were washed three times, and then incubated with pA/G-Tnp pro for 1h at room temperature. After washing, 50 µl TTBL was added into samples and incubated at 37 °C for 1h. Then, added 2 µl 10% SDS and DNA Spike-in into samples incubated at 55 °C for 10 min. Supernatant was transferred into a new tube after putting on magnetic rack for 2 min. 25 µl DNA Extract Beads Pro was added into supernatant incubated 20 min at room temperature, and then washed two times using B&W buffer. After that, DNA Extract Beads Pro was re-suspended in 15 µl ddH₂O, and amplified at 60 °C for 20 cycles. Library quality was examined using Qubit, AATI, and QPCR, and then sequenced using NovaSeq 6000 (Novogene, China). Adaptor and low-quality reads were removed from raw data, and clean data was used for further analysis. Reads were mapped to mouse reference genome mm39 using Bowtie2. Peak calling was performed using MACS2 at *q*-value <0.05. Then, peak was annotated into relative gene regions.

Statistical analysis

Average data are presented as mean ± SE (standard error), and the statistical difference was calculated using two-tail independent-samples *t* test. Methylation level was presented as percentage, and the statistical difference was calculated using Chi-square test. If *p* value < 0.05, the statistical difference was considered to be significant.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (31872312), National R&D Program of China (2022YFC2703500), the Breeding Plan of Shandong Provincial Qingchuang Research Team (Innovation Team of Farm Animal Cloning 012–1622001), and the Doctor Foundation of Qingdao Agricultural University (6631116008).

Conflicts of interest statement

There are no conflicts of interest to declare.

Data availability

The raw data of sequencing is submitted to the database BGI Sub with No. CRA011654.

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Reviewer #1 (Public Review):

With socioeconomic development, more and more people are obese which is an important reason for sub-fertility and infertility. Maternal obesity reduces oocyte quality which may be a reason for the high risk of metabolic diseases for offspring in adulthood. Yet the underlying mechanisms are not well elucidated. Here the authors examined the effects of maternal obesity on oocyte methylation. Hyper-methylation in oocytes was reported by the authors, and the altered methylation in oocytes may be partially transmitted to F2. The authors further explored the association between the metabolome of serum and the altered methylation in oocytes. The authors identified decreased melatonin. Melatonin is involved in regulating the hyper-methylation of high-fat diet (HFD) oocytes, via increasing the expression of DNMTs which is mediated by the cAMP/PKA/CREB pathway.

Strengths:

This study is interesting and should have significant implications for the understanding of the transgenerational inheritance of GDM in humans.

Weaknesses:

The link between altered DNA methylation and offspring metabolic disorders is not well elucidated; how the altered DNA methylation in oocytes escapes reprogramming in transgenerational inheritance is also unclear.

<https://doi.org/10.7554/eLife.97507.1.sa2>

Reviewer #2 (Public Review):

This manuscript offers significant insights into the impact of maternal obesity on oocyte methylation and its transgenerational effects. The study employs comprehensive methodologies, including transgenerational breeding experiments, whole genome bisulfite sequencing, and metabolomics analysis, to explore how high-fat diet (HFD)-induced obesity alters genomic methylation in oocytes and how these changes are inherited by subsequent generations. The findings suggest that maternal obesity induces hyper-methylation in oocytes, which is partly transmitted to F1 and F2 oocytes and livers, potentially contributing to metabolic disorders in offspring. Notably, the study identifies melatonin as a key regulator of this hyper-methylation process, mediated through the cAMP/PKA/CREB pathway.

Strengths:

The study employs comprehensive methodologies, including transgenerational breeding experiments, whole genome bisulfite sequencing, and metabolomics analysis, and provides convincing data.

Weaknesses:

The description in the results section is somewhat verbose. This section (lines 126~227) utilized transgenerational breeding experiments and methylation analysis to demonstrate that maternal obesity-induced alterations in oocyte methylation (including hyper-DMRs and hypo-DMRs) can be partially transmitted to F1 and F2 oocytes and livers. The authors should consider condensing and revising this section for clarity and brevity.

There is a contradiction with Reference 3, but the discrepancy is not discussed. In this study, the authors observed an increase in global methylation in oocytes from HFD mice, whereas Reference 3 indicates Stella insufficiency in oocytes from HFD mice. This Stella insufficiency should lead to decreased methylation (Reference 33). There should be a discussion of how this discrepancy can be reconciled with the authors' findings.

<https://doi.org/10.7554/eLife.97507.1.sa1>

Reviewer #3 (Public Review):

Summary:

Maternal obesity is a health problem for both pregnant women and their offspring. Previous works including work from this group have shown significant DNA methylation changes for offspring of obese pregnancies in mice. In this manuscript, Chao et al digested the potential mechanisms behind the DNA methylation changes. The major observations of the work include transgenerational DNA methylation changes in offspring of maternal obesity, and metabolites such as methionine and melatonin correlated with the above epigenetic changes. Exogenous melatonin treatment could reverse the effects of obesity. The authors further hypothesized that the linkage may be mediated by the cAMP/PKA/CREB pathway to regulate the expression of DNMTs.

Strengths:

The transgenerational change of DNA methylation following HFD is of great interest for future research to follow. The metabolic treatment that could change the DNA methylation in oocytes is also interesting and has potential relevance to future clinical practice.

Weaknesses:

The HFD oocytes have more 5mC signal based on staining and sequencing (Fig 1A-1F). However, the authors also identified almost equal numbers of hyper- and hypo-DMRs, which raises questions regarding where these hypo-DMRs were located and how to interpret their behaviors and functions. These questions are also critical to address in the following mechanistic dissections as the metabolic treatments may also induce bi-directional changes of DNA methylation. The authors should carefully assess these conflicts to make the conclusions solid.

The transgenerational epigenetic modifications are controversial. Even for F0 offspring under maternal obesity, there were different observations compared to this work (Hou, YJ., et al. *Sci Rep*, 2016). The authors should discuss the inconsistencies with previous works.

In addition to the above inconsistencies, the DNA methylation analysis in this work was not carefully evaluated. Several previous works were evaluating the DNA methylation in mice

oocytes, which showed global methylation levels of around 50% (Shirane K, et al. PLoS Genet, 2013; Wang L., et al, Cell, 2014). In Figure 1E, the overall methylation level is about 23% in control, which is significantly different from previous works. The authors should provide more details regarding the WGBS procedure, including but not limited to sequencing coverage, bisulfite conversion rate, etc.

<https://doi.org/10.7554/eLife.97507.1.sa0>